

Introduction

Injected medicines are commonly used in healthcare settings for the prevention, diagnosis, and treatment of various illnesses. Unsafe injection practices put patients and healthcare providers at risk of infectious and non-infectious adverse events and have been associated with a wide variety of procedures and settings. This harm is preventable. Safe injection practices are part of Standard Precautions and are aimed at maintaining basic levels of patient safety and provider protections. As defined by the World Health Organization, a safe injection does not harm the recipient, does not expose the provider to any avoidable risks and does not result in waste that is dangerous for the community.

Purpose

To define and describe practices necessary to safeguard patients and healthcare workers (HCW) from the transmission of infection due to unsafe injection practices.

Scope

This policy must be followed by all healthcare workers working including but not limited to doctors, nurses and pharmacists at Royal Hospital. It also applies to all students and trainees in different departments who are involved in preparing and or delivering injectable medications and performing point of care testing.

Definitions

Safe injection practices:

These are a set of measures to perform injections in an optimally safe manner for patients, healthcare providers, and others.

Aseptic Technique:

Aseptic technique refers to the use of various barriers and precautions to prevent the transfer of microorganisms from HCW and the environment to the patient during a procedure. Sterile is the absence of all microbes.

Multi-dose Vial (MDV):

A multi-dose vial is a vial of liquid medication intended for parenteral administration (injection or infusion) that contains more than one dose of medication. Multi-dose vials are labelled as such by the manufacturer and typically contain an antimicrobial preservative to help prevent the growth of bacteria. The preservative has no effect on viruses and does not protect against contamination when healthcare personnel fail to follow safe injection practices.

Single Dose Vial (SDV)

A single-dose or single-use vial is a vial of liquid medication intended for parenteral administration (injection or infusion) that is meant for use in a single patient for a single

case/procedure/injection. Single-dose or single-use vials are labelled as such by the manufacturer and typically lack an antimicrobial preservative.

Roles and Responsibilities:

- It is the responsibility of Infection Prevention & Control doctor and Infection Control Practitioners to oversee the implementation, monitor adherence to and deliver continuous education of this policy.
- The nurse in charge of each ward is responsible of enforcing the implementation of this policy and ensuring that all staff involved in injectable medications and point of care testing have the knowledge and the competency to carry out such procedure.
- The lead pharmacist is responsible for enforcing the implementation of this policy in his/her department
- All HCW are responsible for adhering to safe injection, infusion, medication vial, and point-of-care testing practices.
- All HCW are responsible to report non-compliance to this policy to infection prevention and control department, nursing department and or hospital administration.

Other related policies:

This policy outlines the standards and practices to be followed that are more detailed as per each subject to the following policies

- Aseptic technique (**POL/IPC/19/Ver.1.0**)
- Blood Glucose monitoring policy (**SOP/NUR/217/Vers.1.0**)
- Converting Drug administration from IV to Oral policy (**PH-POL-2-Ver.1.0**)
- Antibiotic IV to Oral Switch Criteria (**PH-POL-1-Vers.1.0**)
- Multi-Dose/ Single Dose Vials policy (**PH-POL-11-Vers.1.0**)
- Peripheral line insertion & maintenance (**POL/IPC/25/Ver.1.0**)
- Management of needle stick injury (**POL/IPC/16/Ver.2.0**)

To ensure safe injection practices in the hospital, adherence must be followed to the following safe injection, infusion, medication vial, and point-of-care testing practices:

1. Prescribing of Injectable Medicines:

- Medicines should be given by injection only when the practicality and appropriateness of other routes of administration have been excluded. The use of this route should be regularly reviewed in favour of switching to oral administration as soon as clinically appropriate.

2. ASEPTIC TECHNIQUE:

- Perform hand hygiene (clean hands with alcohol based hand sanitizer or with soap and water) before accessing supplies, handling vials and IV solutions,

preparing or administering medications, and conducting point-of-care testing (e.g., blood glucose, coagulation studies, etc.).

- Use aseptic technique in all aspects of parenteral medication preparation, administration, medication vial use, injection, and point-of-care testing.
- Use a mask to contain respiratory droplets when preparing and injecting solution into an intracapsular space (joint), the spine and during lumbar puncture.
- Store, access and prepare medications and supplies in a clean area on a clean surface.
- Avoid having nonsterile contact with sterile areas of devices, containers and drugs.
- Following an emergency event, discard all opened or needle-punctured vials of sterile parenteral products, IV solutions, and single-use containers, such as bags, bottles, syringes.
- Never store needles and syringes unwrapped because sterility cannot be ensured. Keep bulk unwrapped syringes in the original package (e.g., intradermal syringes).
- Place only pre-filled flush syringes (e.g., saline, heparin) that are terminally sterilized by the manufacturer *after* packaging onto a sterile field immediately after opening.
- Disinfect the rubber stopper of medication vials and the neck of glass ampules with sterile 70% alcohol before inserting a needle or breaking the ampule.
- Disinfect catheter hubs, needleless connectors, and injection ports before accessing. Vigorously apply mechanical friction with chlorhexidine/alcohol or sterile 70% isopropyl alcohol, or other approved disinfectant swab. Allow adequate dry time (unless directed otherwise by manufacturer's instructions) before entry.
- Change port protectors as directed per manufacturer's recommendations.
- Never pool left over parenteral medications (vials or IV solutions) for later administration.
- Do not use prefilled syringes to further dilute medication for administration. This is an unsafe practice due to potential for contamination, dosing errors, drug diversion and needlestick injuries. For drugs that require further dilution prior to administration, whether prepared by a pharmacist or a nurse follow the standards in detailed in this policy.

3. TRANSPORTING MEDICATIONS

- Never transport medication filled syringes/needles in pockets or clothing.
- Follow pharmacy protocol for transporting injectable medications.

4. IV SOLUTIONS

- Use an IV solution (e.g., bag, bottle) for only one patient, and then discard.
- Use needleless spiking devices to remove fluid from IV bottles/bags and vials and use for only one patient.

- Never use a container of IV solution (e.g., bag, bottle) to obtain flush solutions for more than one patient.
- Never use infusion supplies, such as needles, syringes, and administration sets, for more than one patient.
- Use needle free systems for all aspects of parenteral medication administration and transfer of solutions between containers.
- Avoid removing closed system transfer devices used for chemotherapy administration once attached. If a second medication needs to be administered, the device should remain on the port, and be flushed before connecting the second medication.

5. FLUSHING

- Use single-use containers for flush solutions, whenever possible.
- If a multidose vial must be used, use it for only one patient and then discard it whenever possible. A multidose vial can be used for more than 1 patient but it must be prepared and stored in the medicine preparation area away from patient's area. Use a new, unused sterile needle and new, unused sterile syringe for each entry into the vial.

6. SYRINGES AND NEEDLES

- Remove needle, cannula, syringe and/or accessory items from sterile packaging immediately before use.
- Avoid using Pharmacy Bulk Packages of sterile unwrapped syringes whenever possible. These bulk packages are primarily intended for use under an ISO Class 5 environment for medication compounding. If used for a mass immunization clinic or allergy testing, open a new, sterile package and discard any remaining syringes at the conclusion of the activity. Do not save for later use.
- Do not use prefilled syringes to further dilute medication for administration. This is an unsafe practice due to potential for contamination, dosing errors, drug diversion and needle stick injuries. For drugs that require further dilution prior to administration, pharmacy personnel should prepare and dispense the diluted formulation in syringes or mini bags whenever possible or dispense single-use vials of the drug and diluent together.
- Do not prepare medication in one syringe to transfer to another syringe. This is not the best practice but if to be used, follow hand hygiene and aseptic technique strictly. Draw up the medication in a clean environment away from patient's clinical area. Be careful not to touch the syringe's plunger while transferring.
- Never withdraw medication from a manufacturer prefilled syringe barrel (carpuject style syringe barrel).
- Never use a syringe for more than one patient even if the needle has been changed between patients.
- Use a new sterile syringe and a new sterile needle for each entry into a vial or IV bag.
- Utilize sharps safety devices (needles/syringes) to administer injections whenever possible.

- Discard syringes, needles, and cannulas in an approved sharps container/receptacle immediately after use and at the point of use.
- Draw up medication into a syringe as close to administration time as feasible. Inject within 1 hour (or as soon as feasible) after drawing up the medication.
- Label all syringes containing medication if not immediately administered. Include patient identification information, names and amounts of all ingredients, and the name or/initials of the person who prepared it, date and time it was prepared, and beyond use date and time.

7. MEDICATION VIALS

- Always follow the manufacturer's instructions for storage and use.
- Check the manufacturer's expiration date on all medication vials prior to use.
- Inspect vials and discard if sterility is known or suspected to be compromised. Examine vials for particulate matter, discoloration, or turbidity; if present, do not use and discard immediately.
- Read the vial label carefully. Vial size does not indicate whether or not a vial is single-use or multidose.
- Store vials with same colored labels and/or same medication with different dosages separately.
- Disinfect the rubber septum on all vials prior to each entry, even after initially removing the cap of a new, unused vial.
- Always use a new sterile syringe and new needle/cannula when entering any vial. Never enter a vial with a syringe or needle/cannula that has been previously used.
- Use single-use or single-dose vials or ampules whenever possible and discard after use on one patient.
- Use multidose medication vials for one patient whenever possible. The risk of viral hepatitis transmission posed by multidose vials has been clearly demonstrated and mandates a practice of using one vial per one patient whenever possible. Infection transmission risk is reduced when multidose vials are dedicated to one patient.
- Store and access multidose vials away from the immediate patient care environment and always use a sterile syringe and needle/cannula each time the vial is accessed.
- Never leave a needle in the septum of a medication vial for multiple medication draws. This provides a direct route for microorganisms to enter the vial and contaminate the fluid.
- Never use a decapping device to remove the top from a vial (e.g., to pour medications). Draw solutions through the diaphragm with a sterile syringe and sterile transfer device or needle using aseptic technique.
- Use needleless transfer devices when reconstituting drugs. Discard transfer device with the vial at the end of the transfer.

- Use a filter needle or filtered transfer device to draw medications from an ampule into a syringe to prevent glass shard and/or potential microbial contamination (currently is not available).
- Never pool or combine leftover contents of vials for later use.
- Discard any vial that has been placed on a known or visibly contaminated surface or a used procedure tray.
- Label a multidose vial with a beyond-use-date when first accessing it. The beyond-use-date after initially entering a multidose vial is 28 days, unless otherwise specified by the manufacturer. The beyond-use date must never be after the manufacturer specified expiration date.
- Check both the beyond-use-date and the manufacturer's expiration date prior to using an opened multidose vial.
- Use multidose *vaccine* vials before the vial expiration date or as noted in the package insert.
- Discard any vials that were used to draw two or more medications into a single syringe.
- Discard the multidose vial with a needleless vial access device after use with a patient.

8. POINT-OF-CARE TESTING (e.g., BLOOD GLUCOSE, COAGULATION STUDIES)

A.LANCETS

- Use single-use, auto-retracting lancing devices for each patient.
- Dispose of all capillary tubes and sharp devices in a sharps container immediately after use and at the point-of-care.
- Never use a finger stick device (e.g., single-use lancets, lancet holding device or pen-like devices that provide multiple lancets in a reloadable cartridge) for more than one patient.
- Never use pen-like lancing devices to blood check glucose.

B. TESTING DEVICES (e.g., BLOOD GLUCOSE METERS, INR METERS)

Whenever possible, blood glucose meters should be assigned to an individual person and not be shared.

If blood glucose meters must be used on more than one patient, ensure they are labeled by the manufacturer for multiple patient use and include adequate instructions for disinfection of the meter between patients.

Clean and disinfect multiple patient use meters (specifically for blood borne pathogens and other infectious agents) after each patient use, using manufacturer recommendations. If the manufacturer does not provide instructions for cleaning and disinfection, then the testing meter should not be used for more than one patient.

Avoid handling test strip containers with soiled gloves to avoid contamination. If a new test strip is needed, discard soiled gloves and perform hand hygiene before obtaining a new test strip.

Clean visible blood and dirt from meters before disinfecting.

Use hospital approved disinfectant and follow manufacturer's contact time when disinfecting meters between patients

Provide training and oversight for HCW that are responsible for conducting point-of-care testing. Conduct training and competency testing at the time of hire and regularly thereafter.

9.BLOOD GLUCOSE MANAGEMENT

A.MULTIDOSE INSULIN VIALS

- If multidose vials of insulin are used, dedicate each to only one patient.

B. INSULIN PENS

- Provide training and oversight on the use of insulin pens to assure competency and use of proper infection prevention practices.
- Dedicate insulin pens for use with only one patient. Never use insulin pens for more than one person. Do not use unassigned or unlabelled insulin pens.
- Affix patient label directly to the insulin pen.
- Label the pen only, not an outer bag, which could contain the incorrect pen; HCP could falsely verify the contents by the label on the bag.
- In healthcare settings, use only single-use, auto-retracting safety needles with insulin pens.
- Dispose of auto-retracting safety needles (e.g., insulin pens) immediately after use at the point-of-care in a sharps container.
- Never store insulin pens with a needle attached.
- Maintain an ample supply of back-up insulin in vials in case of lost or missing pen to discourage resorting to sharing a device.
- Employ workflow supports to prevent the inadvertent use of an insulin pen on more than one patient.
- Store properly labelled insulin pens in secure area with limited access.
- If barcoding is used at the facility, individually barcode the insulin pens for a specific patient. The manufacturer's barcode should not be used as a default.
- Visible alerts and "hard stops" should be built into the barcode scanning process for insulin pens. This added safety check can confirm that the correct pen is being used for the correct patient before continuing with administration and documentation.

Healthcare Workers (HCW) Safety:

- All HCW should maintain vaccination as per immunisation of healthcare workers policy (**POL/IPC/05/Vers1.0**)
- All HCW must follow prevention of needle stick injuries as detailed in Management of needle stick injuries and incidents involving exposure to blood and body fluids(**POL/IPC/12/Vers.2.0**)
- All HCW are required to report exposure to blood and/or body fluids as per Management of needle stick injuries and incidents involving exposure to blood and body fluids (**POL/IPC/12/Vers.2.0**)

- All HCW who prepare or administer injections or other parenteral medications or perform point of care testing must be competent to perform these tasks.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
APIC position paper: safe injection, infusion, and medication vial practices in health care	Association of Professionals in Infection Control	2016	1-18
SAFE INJECTION PRACTICES	Aurora Health Care System Administrative and Clinical Manual	November 2013	1-3
2007 Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Healthcare Settings	CDC	2007	79-80
Policy For The Safe Management Of Injectable Medicines (Adults And Children)	Portsmouth Hospital, NHS trust	June 2016	21

Review / Revision History

Revision History			
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Zaina AL Maskari		Dr. Omaira Al Omairi	Dr. Amal Al Tai